



Concurrency Safety

lockForUpdate()
Row-level locking

DbRetry service
3 attempts, 50ms

Database transactions
ACID guarantees

Invariants

$\text{reserved_count} \leq \text{stock}$
No overselling

Hold expiry is idempotent
Safe to retry

Idempotency

payment_events.idempotency_key
UNIQUE constraint

processed_at timestamp
Double-processing
prevention

Webhook processing is
idempotent
Duplicate-safe

